From Part Prompts to Interaction Poses: Toward Stable Grasping in Door Opening

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Abstract: Successful door motion does not guarantee a stable handle grasp. In our ten-start development pilot, pretrained MolmoBot opened nine doors, but only two runs satisfied the grasp-retention criterion. Our goal is to improve opening with a stable grasp over this baseline. Because MolmoBot already receives a target-part point prompt, we propose converting part information into an explicit interaction pose with a defined role in execution. Currently, we supply privileged oracle poses to test the value of accurate grasp targets. The final method will predict TCP position, orientation and gripper configuration from images, robot state and the existing point prompt, a part mask, or both. A point-conditioned pose-head prototype is implemented and passes a small Train-only integration test; full training, mask conditioning and the final Test comparison remain planned.

Keywords: Part-aware manipulation, grasp stability, interaction poses

1 Introduction

Our objective is to improve stable handle grasping during door opening over pretrained MolmoBot. The local baseline gap between ordinary opening and grasp-retention success motivates establishing a suitable handle grasp and retaining it while opening. These pilot observations are not benchmark-wide estimates, and failing the criterion does not prove physical grasp loss in every case.

We initially considered supplying handle masks to MolmoBot in MolmoSpaces [1, 2]. However, the native door-opening configuration already supplies a target-part image point with an initial conditioning image. A mask may add shape and extent, but its incremental value must be tested. Existing localization does not fully specify gripper orientation, closure or a pose suited to maintaining contact as the door moves.

We propose a supervised tool-center-point (TCP) pose and gripper configuration that a local execution mechanism explicitly consumes. This gives part inputs a measurable control role; supplying a point or mask alone does not establish effective use. The work has two stages. First, our current oracle study supplies expert-derived poses to test whether accurate targets help. Second, an implemented pose-head prototype will be trained to predict those targets from part information and current observations, then use the same execution mechanism. The final goal is improved stable-grasp opening over the baseline while preserving ordinary opening success.

2 Related Work

Simulation and pretrained policies. MolmoSpaces provides indoor simulation assets, data-generation tools and benchmarking infrastructure [1]. MolmoBot learns manipulation from large-scale synthetic demonstrations and supports optional image-point conditioning [2]. We use the released RB-Y1 multitask policy and MuJoCo environment, adding grasp-stage guidance rather than training a complete policy from scratch.

From local affordances to geometric actions. Where2Act connects articulated-part actionability to contact-conditioned interaction parameters, including gripper orientation [3]. PerAct predicts structured 6-DoF manipulation actions from language and RGB-D observations [4]; Act3D predicts end-effector keyposes using a spatial feature representation [5]. These works motivate representing how to interact, in addition to where. Our approach preserves MolmoBot for full-task behavior and studies whether part-conditioned pose guidance improves stable grasping during door opening.

3 Problem Definition

At time t, the native policy observes camera images I_t, robot state s_t, task instruction ℓ, and native target-point/conditioning-image input p. It produces a chunk of joint-space actions. The proposed pose module additionally uses part evidence z_t: the existing point prompt, a part mask, or both. The final learned mapping is $\hat{y}_t = f_\theta(I_t, s_t, \ell, z_t)$, with $\hat{y}_t = (\hat{T}_G^{B_t}, \hat{c})$, where $\hat{T}_G^{B_t} \in SE(3)$ is a desired TCP pose in the current robot-base frame and $\hat{c}$ specifies gripper configuration. The reported control pilots use supplied oracle targets. The new predictor has undergone only a four-update integration test, rather than the planned full training run.

Data. We prepared 5,004 pose references from official Train demonstrations across 977 contributing houses. The early-grasp view retains 4,866 targets and 19,464 observation–target pairs. We use official Train for learning and official Val as Test, with 187 currently available early-grasp targets. Selection requires source grasp flags and measured closure before 10% opening, with no prior major opening. Four preceding observations share each target in their respective robot-base frames. These are recorded expert TCP references; full-task grasp retention is not certified by the source labels.

Objective and evaluation. Let S_{open} denote reaching the configured opening fraction of 0.67 and S_{grasp} denote acquiring a qualifying grasp before substantial opening and retaining it through completion. We aim to increase $J_{\text{stable}}(\pi) = \Pr_\pi(S_{\text{open}} \wedge S_{\text{grasp}})$ over frozen MolmoBot, also reporting ordinary opening, grasp acquisition and pose error. Retention diagnostics sample forces and handle-relative TCP motion at 10 Hz; 2 cm/10-degree limits are proxies under validation. Comparisons share starts, seeds and action budgets, disclosing additional sensing. We separate Train/Test houses; Test never selects settings or checkpoints, and prior inspections are recorded. The existing 197 expert trajectories form the Test base. Verification and expansion toward 400 total cases are deferred; candidate identities do not establish qualification. This split does not prove absence from MolmoBot pretraining.

4 Technical Approach

Stage 1: supplied oracle poses. In the full-start policy pilot, we supply an expert-derived grasp pose and apply bounded corrections around handle contact. This tests whether a more specific target helps beyond a grasp guard alone, before training a predictor. MolmoBot still generates approach and opening actions; expert action sequences are not replayed in this comparison.

Stage 2: implemented pose-head prototype. A frozen DINOv2 encoder extracts global and point-local features from three current RGB views. A three-layer transformer combines them with robot joints and push/pull context, predicting TCP translation, rotation, active arm and finger-displacement span. Only the added head is trained. Current-frame simulator handle points are an additional input beyond MolmoBot's initial-image prompt; this distinction must be controlled in comparisons. Missing points have explicit visibility flags. Masks remain a future input extension. Expert poses and arm labels supervise training but are not inference inputs.

Supervised pose target. For a recorded grasp pose $T_{G^*}^W$, the target at an earlier observation is $T_{G^*}^{B_t} = (T_{B_t}^W)^{-1} T_{G^*}^W$. The prototype uses Smooth L1 for translation and finger span, squared rotation matrix distance, and cross-entropy for arm selection. Finger span is a measured displacement proxy; native execution still uses guarded open/close commands.

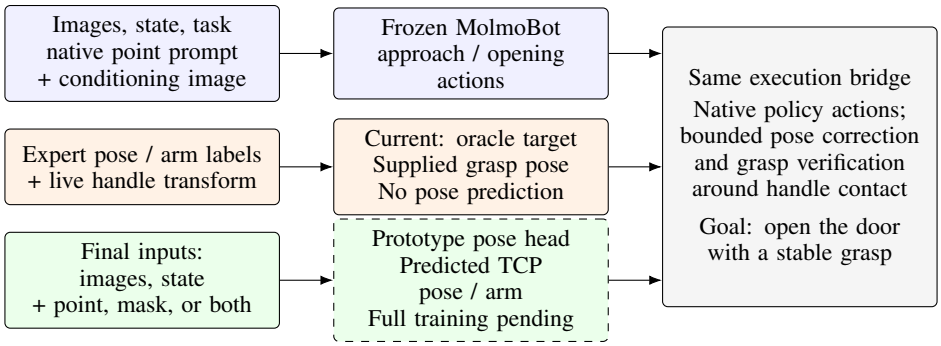


Figure 1: Oracle targets (orange) and the pose-head prototype (green) feed the same execution bridge. Full training and mask inputs remain planned.

Execution mechanism. The bridge converts bounded local corrections into native relative commands, verifies contact and refreshes policy actions after intervention. Oracle targets use expert handle-relative poses and live handle transforms. Initial alignment alone does not ensure retention; opening also requires feedback.

Controls and training plan. Compare frozen MolmoBot, a grasp guard, oracle correction and learned pose guidance. Point/mask ablations will test input dependence. Full training is pending: 20,000 optimizer updates, effective batch four and Train diagnostics every 500 updates. The final checkpoint is fixed before Test evaluation.

5 Preliminary Results

Local grasp acquisition: 7/10 to 10/10. We first isolated grasping by replaying the expert approach, then executing a local controller with an expert-derived TCP target. On the same ten Train development cases, a wider closing-position tolerance (20 mm rather than 12 mm) and bounded base recovery improved sustained bilateral contact from 7/10 to 10/10 (Table 1). Each controller had 120 decisions (12 seconds), with privileged expert targets and simulator handle/contact state. This measures local acquisition after an expert approach, not full-start opening or force-qualified opening retention.

Table 1: Retained local grasp comparison after expert approach. Success requires bilateral finger contact throughout the final second of the 120-decision test.

Controller	Sustained local contact
Original 12 mm closing gate	7/10
20 mm closing gate + bounded base recovery	10/10

Three workers finished simulation but failed a 10^{-8} m reference check. Post-hoc review confirmed identical discrete outcomes and trajectory differences below 0.1 mm/0.05 degrees. Original error statuses and success labels are preserved.

Full-start MolmoBot comparison. In a separate ten-start Train pilot, MolmoBot generates approach and opening actions from the original closed-door states. Table 2 shows the baseline gap: 9/10 ordinary openings versus 2/10 retained-grasp openings, our primary improvement target.

All 30 rollouts matched initial inputs and first policy proposals. Retained-grasp opening rises from 2/10 to 3/10 with the guard and 4/10 with oracle correction, but oracle ordinary opening falls to 8/10. This tradeoff does not establish overall improvement or learned-model performance.

Table 2: Completed ten-start Train development pilot, seed 42, 400-action budget. Retained-grasp values use the historical pilot criterion.

Method	Opening threshold	Retained-grasp opening
Frozen MolmoBot	9/10	2/10
MolmoBot + grasp guard	10/10	3/10
Guard + oracle pose correction	8/10	4/10

Data and training readiness. The source data and early-grasp indices are prepared. The new pipeline extracted real DINOv2 features from four Train observations of one trajectory, completed four optimizer updates, saved and reloaded a checkpoint, and produced an offline Train pose-error report without accessing Test. This verifies implementation, not learned grasping performance. No full 20,000-update training or learned-policy GPU evaluation has been run.

Supporting diagnostics. Fractional mask support preserved all 21 visible handles in saved preprocessing checks; a 50%-foreground patch rule erased 15. Three expert command replays opened 3/3 doors in each of two noise modes, but a newer onset-based retention metric passed only 1/3 and 0/3. Those rates use different semantics from Table 2; a declared ten-step continuation was added after source-recorded success to observe holding. These checks motivate further metric and controller-compatibility review rather than certifying the entire Test set.

Next, we will train for 20,000 updates and compare predicted poses with frozen MolmoBot on matched starts. Test verification/expansion toward 400 cases is deferred; action-candidate scoring remains a separate optional direction.

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A GenAI Use

Generative AI (OpenAI Codex) helped me analyze the dataset, inspect robot motions, and identify failure cases and possible causes. When I asked about research concepts relevant to this project, it suggested related papers for me to explore. It also helped refine the organization, clarity and wording of this report.